

Relative Exponential Periods on Polynomial Flags: Polynomial Boundary Systems and an Elliptic Comparison Theorem

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Abstract

For a compact polynomially parametrized two-chain D and a polynomial phase q , we construct the differential system of the actual integrals $\int_D e^{zq} \omega$ using polynomial representatives in twisted algebraic de Rham cohomology. Its forcing terms are the actual integrals on the fixed faces of D . No spatial denominators or independently normalized punctual periods are introduced. A separate actual-solution criterion upgrades nonmembership in the rational linear span of the boundary functions to algebraic independence over their fraction field, using projective differential-Galois disjointness and a special-linear affine orbit.

For the triangle with vertices $(-1/2, -1), (1/2, 0), (-1/2, 1)$ and phase $q = y^2 - x^3 + 3x$, we prove that the two moments with amplitudes 1 and x are algebraically independent over the actual face-function field. Their values are algebraically independent over the corresponding face-value field at every nonzero algebraic parameter. Every polynomial amplitude has a unique two-coordinate remainder and an explicit polynomial twisted Stokes certificate. The geometric nonvanishing is proved on the actual density by continuation to the two nodal fibers, and the projective exclusion is proved from the explicitly identified cubic boundary block. The actual results do not use the compact polynomial-line completeness theorem; that theorem supplies only the final formal interpretation of relations among the boundary symbols.

We retain the face-lattice comparison, rooted Picard–Lefschetz theorem, and denominator-free single-phase comparison on an explicitly observable locus. For more general collections we give an abstract local specialization theorem with separate generic-completeness and flatness hypotheses. Numerical injectivity is stated on the specialized fiber, not on the unspecialized local functional algebra. General isotypic and multicolumn geometric comparisons are not asserted without their actual linear and determinant-line realizations.

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1 Introduction and scope

Put $k = \overline{\mathbb{Q}}$ and $k_{\mathbb{R}} = k \cap \mathbb{R}$. We consider integrals

$$I_D(\omega)(z) = \int_D e^{zq} \omega$$

over compact polynomially parametrized two-chains. The polynomial phase may be complex in the algebraic construction; real phases are imposed only where a real one-dimensional phase density is used.

A complete comparison must keep three questions separate. Polynomial reduction constructs a boundary-forced system. Differential Galois theory classifies polynomial relations of an actual solution of that system. Arithmetic lifting transfers a functional classification to algebraic parameters. None of these questions is answered solely by the list of composition factors of a holonomic direct image.

1.1 The polynomial boundary construction

Over $F_0 = k(z)$, put

$$d_z = d + z dq \wedge, \quad H_q^2 = \Omega_{F_0[x,y]/F_0}^2 / d_z \Omega_{F_0[x,y]/F_0}^1.$$

Theorem 10.1 proves that this space has a basis represented by z -independent polynomial forms ω_i and identities

$$q\omega_i = \sum_j A_{ij}(z)\omega_j + d_z \beta_i.$$

Here β_i is polynomial in x, y and rational in z . Integrating the identities on the original chain gives

$$U' = AU + b, \quad U_i = \int_D e^{zq} \omega_i, \quad b_i = \int_{\partial D} e^{zq} \beta_i.$$

The boundary field is generated by all polynomial-amplitude integrals on these *fixed faces* and their endpoint exponentials. It is not the field of all compact polynomial-line periods.

Theorem 11.2 supplies an actual-solution comparison when the homogeneous group contains SL_n , its adjoint module is excluded from the boundary tensor category, and the actual column has nonzero quotient by the finite rational linear boundary space. These are separate hypotheses; abstract nonzero cohomology alone does not imply the last one.

1.2 The explicit elliptic theorem

Let

$$q = y^2 - x^3 + 3x, \quad D = \{-1/2 \leq x \leq 1/2, |y| \leq 1/2 - x\}, \quad M_j = \int_D x^j e^{zq} dx dy.$$

Set

$$\begin{aligned} p(x) &= -x^3 + x^2 + 2x + \frac{1}{4}, & P_j(z) &= \int_{-1/2}^{1/2} x^j e^{zp(x)} dx, \\ Q_0(z) &= \int_{-1}^1 e^{z(y^2 - 11/8)} dy, & E_a(z) &= e^{-3z/8}, \quad E_b(z) = e^{11z/8}. \end{aligned}$$

Then the actual face field is

$$K_{\partial,0} = k(z)(P_0, P_1, Q_0, E_a, E_b).$$

Theorems 15.2 and 16.1 prove

$$K_{\partial,0}[M_0, M_1] \simeq K_{\partial,0}[X_0, X_1]$$

and, for every $\xi \in k^\times$,

$$K_{\partial,\xi}[M_0(\xi), M_1(\xi)] \simeq K_{\partial,\xi}[X_0, X_1],$$

where $K_{\partial,\xi}$ is the field of the corresponding five boundary values. Every polynomial amplitude reduces to these two coordinates plus actual face terms. In particular, its value is algebraic over the face-value field exactly when its specialized polynomial twisted de Rham class is zero (Corollary 16.2).

The proof has an explicit chain. Polynomial identities give the actual rank-two system. Rational Gelfand–Leray certificates identify its phase Gauss–Manin matrix. A nonzero nodal variation of the actual density excludes the rational linear boundary span. The boundary group has only one nonabelian simple factor, whose adjoint exponential factors differ from those of the elliptic system. Finally the affine orbit is dense and the combined E -function system is ordinary at every nonzero finite parameter. No determinant-line generator is needed.

1.3 Geometry, formal relations, and the general boundary

The geometric part retains polynomial relative exactness on the observable locus of Definition 7.1, using the relative homology sequence, rooted Picard–Lefschetz variation, and Bonnet’s polynomial relative-exactness theorem. The disjoint-block comparison is stated on the actual common phase cover where its hypotheses are checked. Rational Jacobian morphisms and complex linear combinations of them are carefully distinguished.

The formal interpretation uses the period operations of Kontsevich–Zagier [11] and the compact polynomial-line comparison of Papers II–III [13, 14]; related exponential-period realizations are discussed in [4]. These inputs concern boundary identities. The actual independence theorem for the elliptic triangle is independent of their formal completeness.

For a general collection, Theorem 18.1 isolates the local arithmetic step. It assumes an explicit generically complete formal ideal and flatness over a boundary DVR. We also retain the abstract affine special-linear orbit calculation and a unit-normalized flat model. Their application to arbitrary flag collections requires actual linear comparisons and, at full column rank, formally realized determinant-line maps. No automatic equivalence with a condition on the original relative monodromy cocycles is asserted.

Part I

Geometric functional comparison

2 Polynomial flags and phase densities

2.1 Polynomial flag chains

Definition 2.1 (Polynomial two-chain). A polynomial two-chain over $k_{\mathbb{R}}$ is a finite k -linear combination of chains

$$D = \Phi_*[0, 1]^2, \quad \Phi \in k_{\mathbb{R}}[u, v]^2.$$

Its boundary is the alternating sum of the four polynomial face chains. A polynomial flag chain is a polynomial two-chain for which, after an affine change of the square coordinates,

$$\Phi(u, v) = (X(u), A(u) + vB(u))$$

with $X, A, B \in k_{\mathbb{R}}[u]$.

The chain language is preferable to set-theoretic domains: it records orientation, multiplicity, and algebraic change of variables without requiring injectivity of Φ .

For a polynomial two-form Ω , define

$$F_{D,q,\Omega}(z) = \int_D e^{zq} \Omega.$$

Pulling back to the square writes this as

$$F_{D,q,\Omega}(z) = \int_0^1 \int_0^1 P(u, v) e^{zQ(u,v)} du dv, \quad P, Q \in k[u, v]. \quad (2.1)$$

2.2 Gelfand–Leray density

Let $q \in k_{\mathbb{R}}[x, y]$ be nonconstant and let D be a real polynomial flag chain. Put $T = q(|D|) \subset \mathbb{R}$, a finite union of compact intervals. Remove the critical values of q , the critical values of the restrictions of q to the faces, and the phase values of the face vertices. The complement is a finite union of open intervals e .

For $t \in e$, let

$$C_t = q^{-1}(t), \quad B_t = C_t \cap |\partial D|, \quad \gamma_t = D \cap C_t.$$

The last object is an oriented relative one-cycle in (C_t, B_t) , with the multiplicities inherited from D .

Let Ω be a polynomial two-form. On the smooth locus of q , choose a one-form α satisfying

$$dq \wedge \alpha = \Omega. \quad (2.2)$$

The restriction class of α to C_t is independent of the choice. Define

$$\rho_e(t) = \int_{\gamma_t} \alpha_t. \quad (2.3)$$

Lemma 2.2 (Phase-density formula). *After discarding chain components on which the pullback of Ω is zero, there is an L^1 density ρ on the compact set $T = q(|D|)$ such that*

$$F_{D,q,\Omega}(z) = \int_T e^{zt} \rho(t) dt. \quad (2.4)$$

On every regular interval e , the restriction ρ_e is real analytic and is a branch of a period of the algebraic Gauss–Manin system. In particular, it admits multivalued holomorphic continuation on the regular phase line. No claim that ρ_e is an algebraic function is made.

Proof. On a compact subinterval of e , the map q is a submersion on the sliced chain. The Gelfand–Leray or coarea formula gives

$$\int_{D \cap q^{-1}(e)} e^{zq} \Omega = \int_e e^{zt} \left(\int_{\gamma_t} \alpha_t \right) dt.$$

The inner integral is a period of the relative algebraic de Rham class $[\alpha_t]$ and therefore is real analytic on e and has the usual multivalued Gauss–Manin continuation. The coarea decomposition of the finite measure obtained by integrating the polynomial density on the compact chain shows that the resulting density is integrable across the finitely many critical and boundary values. Summing the regular intervals proves (2.4). A full-dimensional component on which q is constant cannot occur for nonconstant polynomial q ; a lower-dimensional component pulls every two-form back to zero. $\square$

2.3 Laplace uniqueness

Lemma 2.3 (Compact-support Laplace uniqueness). *Let ρ be a compactly supported integrable function on $\mathbb{R}$. If*

$$\int_{\mathbb{R}} e^{zt} \rho(t) dt = 0 \quad (z \in \mathbb{C}),$$

then $\rho = 0$ almost everywhere. If ρ is analytic on each member of a finite interval decomposition, it vanishes identically on every open interval.

Proof. Restriction to $z = is$, $s \in \mathbb{R}$, is the Fourier transform of the compactly supported L^1 function ρ . Fourier uniqueness gives almost-everywhere vanishing, and analyticity gives intervalwise vanishing. $\square$

For multiphase families it is useful to retain the Cauchy-transform form.

Lemma 2.4 (Cauchy jump separation). *Let $T \subset \mathbb{R}$ be a finite union of compact intervals and let ρ_j be piecewise analytic integrable densities on T . If*

$$\sum_j c_j \int_T e^{zt} \rho_j(t) dt = 0 \quad (z \in \mathbb{C}),$$

then

$$\sum_j c_j \rho_j(t) = 0$$

on every open regular interval.

Proof. All moments of the combined density vanish. Its Cauchy transform

$$\mathcal{C}(s) = \int_T \frac{\sum_j c_j \rho_j(t)}{s - t} dt$$

therefore vanishes for large s and hence on $\mathbb{C} \setminus T$. At an interior point of a regular interval, the two lateral values differ by $2\pi i$ times the analytic density. The jump is zero, so the density vanishes. $\square$

Remark 2.5 (Why real phase support is imposed). For a general complex-valued polynomial phase on a real two-chain, the pushforward is a two-dimensional measure in the complex phase plane. Its holomorphic moments can all vanish without the measure vanishing. Thus the one-variable Cauchy jump argument is unavailable. The phase-fibered hypothesis is mathematical content, not a notational convenience.

3 Polynomial-flag periods are E -functions

We use Beukers' normalization of Siegel E -functions. An entire function

$$F(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}$$

is an E -function if $a_n \in k$, the joint logarithmic height of the first n coefficients is $O(n)$, and F satisfies a nonzero linear differential equation over $k(z)$.

Theorem 3.1 (Polynomial-flag E -function theorem). *For $P, Q \in k[u, v]$, the function*

$$F(z) = \int_0^1 \int_0^1 P(u, v) e^{zQ(u, v)} du dv$$

is an E -function. Consequently every polynomial-flag period $F_{D, q, \Omega}$ is an E -function.

Proof. Termwise integration gives

$$F(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}, \quad a_n = \int_0^1 \int_0^1 P(u, v) Q(u, v)^n du dv.$$

Choose a number field K containing the coefficients and an integer $d \geq 1$ such that dP, dQ have algebraic-integer coefficients. The degree of PQ^n is $O(n)$, the number of its monomials is polynomial in n , and every archimedean conjugate of every coefficient is bounded by C^n for a fixed C . Writing

$$PQ^n = \sum_{i, j} c_{i, j, n} u^i v^j$$

gives

$$a_n = \sum_{i, j} \frac{c_{i, j, n}}{(i+1)(j+1)}.$$

If $N_n = O(n)$ bounds all occurring exponents, then

$$d^{n+1} \text{lcm}(1, \dots, N_n + 1)^2$$

clears all denominators. The elementary Chebyshev bound gives $\log \text{lcm}(1, \dots, N) = O(N)$; hence the common denominator has logarithm $O(n)$. The conjugate and monomial-count estimates give the same bound for the joint height.

The integrand $P(u, v) e^{zQ(u, v)}$ is holonomic in (u, v, z) . Holonomic D -modules are stable under algebraic direct image, and integration over the fixed relative chain gives a holonomic function of z ; see [9, Chapters 1 and 3]. Therefore F satisfies a nonzero differential equation over $k(z)$. The integral is entire by uniform convergence on compact subsets. All three E -function conditions hold. $\square$

Theorem 3.2 (Beukers lifting). *Let $F_1, \dots, F_N$ be E -functions forming a vector solution of a first-order system over $k(z)$, and let $\xi \in k^\times$ be a nonsingular point. Every homogeneous polynomial relation of degree d among the values $F_i(\xi)$ lifts to a polynomial-coefficient functional relation, homogeneous of the same degree, specializing to the prescribed relation at $z = \xi$.*

Proof. This is the combination of [2, Theorems 1.3 and 3.2]; affine relations are homogenized by adjoining the constant E -function 1. $\square$

4 Relative affine-Galois linearity

Theorem 4.1 (Kolchin–Ostrowski). *Let K be a differential field of characteristic zero with algebraically closed constants C . Suppose $Y_1, \dots, Y_m$ lie in a no-new-constants differential extension and satisfy $DY_i \in K$. They are algebraically dependent over K if and only if some nonzero constant linear combination of them belongs to K .*

This is the classical antiderivative theorem; see [10, 16].

Let

$$R_Y = \{c \in C^m : c \cdot \mathbf{Y} \in K\}.$$

Theorem 4.2 (Affine-linear presentation). *Let $r = \dim_C R_Y$. There is $S \in \mathrm{GL}_m(C)$ such that, for $\mathbf{Z} = S\mathbf{Y}$,*

$$Z_i = g_i \in K \quad (1 \leq i \leq r),$$

and $Z_{r+1}, \dots, Z_m$ are algebraically independent over K . Hence

$$\ker(K[\mathbf{X}] \rightarrow K[\mathbf{Y}]) = (Z_1 - g_1, \dots, Z_r - g_r).$$

Proof. Choose a basis of R_Y and extend it to a basis of C^m ; use these row vectors to form S . The first r transformed coordinates lie in K . If the remaining coordinates were algebraically dependent, Theorem 4.1 would put a nonzero constant linear combination of them in K , contradicting the choice of complement. Division by the monic linear polynomials $Z_i - g_i$ leaves a polynomial in algebraically independent variables, proving the kernel statement. $\square$

4.1 One iterated layer

Let x be an outer variable and let K be a differential field containing the inner period functions and their x -derivatives. If

$$V_i(x, z) = \int_{x_0}^x f_i(t, z) dt, \quad f_i \in K,$$

then Theorem 4.2 describes the complete polynomial relation ideal among the partial outer integrals over K .

Remark 4.3 (Endpoint evaluation is a separate problem). Take algebraically independent antiderivatives

$$V_i(x) = \int_b^x f_i(t) dt$$

over a differential field K . They have no functional algebraic relation over K , but $V_1(b) = \dots = V_m(b) = 0$. Endpoint evaluation therefore creates relations not present in the variable-endpoint function field. In particular, Theorem 4.2 cannot by itself justify a value-level polynomial theorem for completed two-variable periods.

5 The face lattice and the generalized Jacobian

Let $\bar{f} : \bar{\mathcal{C}} \rightarrow S$ be a smooth proper family of connected curves and let $\pi : B \rightarrow S$ be a nonempty reduced divisor finite étale over S . Define

$$\mathcal{F}_B = \ker \left(\pi_* \mathbb{Z}_B \xrightarrow{\deg} \mathbb{Z}_S \right). \quad (5.1)$$

Proposition 5.1 (Relative homology sequence). *There is a natural exact sequence of integral local systems*

$$0 \longrightarrow R_1 \bar{f}_* \mathbb{Z} \longrightarrow R_1 \bar{f}_*(\bar{C}, B; \mathbb{Z}) \xrightarrow{\partial} \mathcal{F}_B \longrightarrow 0. \quad (5.2)$$

Proof. Apply the homology long exact sequence of the pair $(\bar{C}_s, B_s)$. Connectedness gives $H_0(\bar{C}_s, \mathbb{Z}) = \mathbb{Z}$, and the map $H_0(B_s, \mathbb{Z}) \rightarrow H_0(\bar{C}_s, \mathbb{Z})$ is the degree map. The construction is locally constant in s and therefore yields (5.2). $\square$

Proposition 5.2 (Generalized-Jacobian torus). *Let $J = \text{Pic}^0(\bar{C}/S)$ and $J_B = \text{Pic}^0(\bar{C}/S, B)$. There is an exact sequence*

$$1 \longrightarrow T_B \longrightarrow J_B \longrightarrow J \longrightarrow 0, \quad T_B = \text{Res}_{B/S} \mathbb{G}_m / \mathbb{G}_m, \quad (5.3)$$

and

$$X^*(T_B) \simeq \mathcal{F}_B. \quad (5.4)$$

Proof. For reduced modulus, the generalized Jacobian is the extension of the ordinary Jacobian by the quotient of the Weil restriction of $\mathbb{G}_m$ by the diagonal $\mathbb{G}_m$; see [15, Chapter V]. Characters are contravariant. The character sequence associated to

$$1 \rightarrow \mathbb{G}_m \rightarrow \text{Res}_{B/S} \mathbb{G}_m \rightarrow T_B \rightarrow 1$$

is

$$0 \rightarrow X^*(T_B) \rightarrow \pi_* \mathbb{Z}_B \xrightarrow{\deg} \mathbb{Z}_S \rightarrow 0,$$

which is (5.4). $\square$

Remark 5.3 (Variance and weights). The relative-homology quotient $\mathcal{F}_B$ is a lattice of boundary classes. The same lattice is the character group of the torus T_B ; the torus itself is dual. In mixed-Hodge terms, relative homology and punctured-curve cohomology carry opposite Tate twists. The slogan “the face module is the toric part” is correct only after this variance is specified.

For a polynomial flag slice, B_t is the finite set of intersections with the oriented faces and

$$\partial \gamma_t \in \mathcal{F}_{B,t}.$$

The equation $\deg \partial \gamma_t = 0$ is simultaneously the balanced face relation and the boundary-of-boundary identity.

6 Rooted Picard–Lefschetz observability

Let H be a finite-dimensional vector space and suppose the monodromy group G is generated by relative Picard–Lefschetz transformations

$$(T_i - I)\xi = \varepsilon_i \langle \tilde{\nu}_i, \xi \rangle \nu_i, \quad \varepsilon_i \in \mathbb{Q}^\times. \quad (6.1)$$

Here ν_i and $\tilde{\nu}_i$ are vanishing and dual vanishing cycles. The nonzero coefficient allows both orientations and positive powers of local monodromy after a finite cover. We use the standard relative Picard–Lefschetz formula; see, for example, [8, Chapters 2–3]. Let γ be a relative slice class.

Definition 6.1 (Rooted variation graph). Put

$$I_\gamma = \{i : \langle \tilde{\nu}_i, \gamma \rangle \neq 0\}.$$

Put a directed edge $i \rightarrow j$ when

$$\langle \tilde{\nu}_j, \nu_i \rangle \neq 0.$$

Let R_γ be the vertices reachable from I_γ , and set

$$\mathcal{V}_\gamma = \text{span}\{\nu_i : i \in R_\gamma\}, \quad \mathcal{O}_\gamma = \text{span}\{g\gamma - \gamma : g \in G\}.$$

Theorem 6.2 (Exact rooted orbit theorem). *One has*

$$\mathcal{O}_\gamma = \mathcal{V}_\gamma.$$

The space is invariant under G .

Proof. The space $\mathcal{O}_\gamma$ is invariant because

$$h(g\gamma - \gamma) = (hg\gamma - \gamma) - (h\gamma - \gamma).$$

If $i \in I_\gamma$, (6.1) gives $\nu_i \in \mathcal{O}_\gamma$. If $\nu_i \in \mathcal{O}_\gamma$ and $i \rightarrow j$, then

$$(T_j - I)\nu_i = \varepsilon_j \langle \tilde{\nu}_j, \nu_i \rangle \nu_j,$$

so $\nu_j \in \mathcal{O}_\gamma$. Hence $\mathcal{V}_\gamma \subseteq \mathcal{O}_\gamma$. The same formula shows that every T_j preserves $\mathcal{V}_\gamma$.

Conversely, for every generator,

$$T_j\gamma - \gamma \in \mathcal{V}_\gamma.$$

If $g\gamma - \gamma \in \mathcal{V}_\gamma$, then

$$T_j g\gamma - \gamma = T_j(g\gamma - \gamma) + (T_j\gamma - \gamma) \in \mathcal{V}_\gamma.$$

The same argument applies to T_j^{-1} , since $T_j^{-1} - I$ has the same rank-one image. Induction on word length in the generators and their inverses gives $g\gamma - \gamma \in \mathcal{V}_\gamma$ for every g , proving the reverse inclusion. $\square$

On absolute curve homology the intersection graph is undirected.

Corollary 6.3 (Connected-graph observability). *Suppose the vanishing cycles span V , their intersection graph is connected, and $I_\gamma \neq \emptyset$. Then*

$$\mathcal{O}_\gamma = V.$$

Proof. Every vertex is reachable from the nonempty active set. Apply Theorem 6.2. $\square$

Proposition 6.4 (Absolute irreducibility). *Assume in addition that V carries a nondegenerate alternating pairing and that the T_i are the associated symplectic transvections. Under the hypotheses of Corollary 6.3, the representation on V is absolutely irreducible.*

Proof. Let $K/\mathbb{Q}$ be a characteristic-zero extension and let $0 \neq W \subset V_K$ be invariant. Choose $0 \neq w \in W$. Since the vanishing cycles span and the pairing is nondegenerate, some $\langle w, \nu_i \rangle$ is nonzero. Then $(T_i - I)w$ is a nonzero multiple of ν_i , so $\nu_i \in W$. Connectivity propagates membership to every vanishing cycle. Thus $W = V_K$. $\square$

7 Denominator-free relative exactness

Definition 7.1 (Admissible observable flag datum). An admissible observable flag datum is a triple (D, q, Ω) with D a polynomial flag chain, $q \in k_{\mathbb{R}}[x, y]$, and $\Omega \in \Omega_{k[x, y]/k}^2$, satisfying:

- (A1) q has finite critical locus and every fibre $q^{-1}(t)$ is connected;
- (A2) the generic affine fibre has one place at infinity;
- (A3) after resolving the compactified pencil, every finite singular fibre has exactly one ordinary node and the finite critical values are distinct; over their complement the fibres form a smooth proper curve family, and the face intersections become a finite étale divisor after a finite base change;
- (A4) the resulting Lefschetz vanishing cycles span H_1 of the compactified generic fibre and have connected intersection graph; the relative continuation around these nodes obeys the rank-one formula (6.1), on a common cover labeling the face points;
- (A5) the sliced chain is nonzero on at least one regular phase interval, and on every interval on which it is nonzero it has nonzero incidence with at least one Lefschetz vanishing cycle.

The connected-fibre and finite-critical-locus assumptions put q in the primitive quasi-fibered class of Bonnet. The one-place-at-infinity condition removes an additional puncture lattice: it identifies the first homology of the affine generic fibre with that of its compactification.

Theorem 7.2 (Single-phase denominator-free functional theorem). *Let (D, q, Ω) be an admissible observable flag datum and put*

$$F(z) = \int_D e^{zq} \Omega.$$

If $F \equiv 0$, then there is a polynomial $a \in k[x, y]$ such that

$$\Omega = da \wedge dq.$$

Consequently

$$e^{zq} \Omega = d(e^{zq} a dq)$$

and

$$F(z) = \int_{\partial D} e^{zq} a dq.$$

The boundary relation is a compact polynomial-line functional relation and is therefore generated by the pullback–Stokes operations of [13].

The algebraic step is Bonnet’s theorem that topological relative exactness equals polynomial relative exactness for primitive quasi-fibered polynomial maps [3].

7.1 Bonnet’s theorem

For a polynomial map $q : \mathbb{C}^2 \rightarrow \mathbb{C}$, a polynomial one-form ω is *topologically relatively exact* if its restriction to every generic fibre has zero integral on every loop. It is *algebraically relatively exact* if

$$\omega = dR + a dq, \quad R, a \in \mathbb{C}[x, y].$$

Lemma 7.3 (Bonnet’s hypotheses in relative dimension one). *Let $q : \mathbb{C}^2 \rightarrow \mathbb{C}$ be nonconstant. If its critical locus is finite and every fibre is connected, then q is primitive and quasi-fibered in Bonnet’s terminology.*

Proof. A finite critical locus has codimension two, so q is nonsingular in codimension one. Since the target has dimension one, Bonnet’s requirement that the fibres be 1-generically connected means that every fibre is connected. His 2-generic nonemptiness condition is automatic: for every $c \in \mathbb{C}$, the proper ideal $(q - c) \subset \mathbb{C}[x, y]$ has a zero by the Nullstellensatz. The weaker generic connectedness and nonemptiness conditions defining primitivity follow as well. $\square$

Theorem 7.4 (Bonnet–Gavrilov relative exactness). *If $q : \mathbb{C}^2 \rightarrow \mathbb{C}$ has finite critical locus and every fibre is connected, then every topologically relatively exact polynomial one-form is algebraically relatively exact.*

Proof. Apply Lemma 7.3 and [3, Theorem 1.5]. Bonnet also records the earlier two-variable connected-and-reduced theorem of Gavrilov. $\square$

7.2 From a zero phase density to relative exactness

Let Ω be a polynomial two-form. Choose a polynomial one-form ω with

$$d\omega = \Omega; \tag{7.1}$$

this is possible because $H_{\text{dR}}^2(\mathbb{A}^2) = 0$. On a regular fibre choose α as in (2.2). The Gelfand–Leray formula is

$$\nabla[\omega_t] = [\alpha_t]. \tag{7.2}$$

Lemma 7.5 (Observable density kills the Gelfand–Leray class). *Let (D, q, Ω) satisfy (A2)–(A5) of Definition 7.1. If the phase density vanishes on a regular interval on which the sliced chain is nonzero and has nonzero Lefschetz incidence, then*

$$[\alpha_t] = 0 \in H_{\text{dR}}^1(C_t)$$

for the generic affine fibre.

Proof. On the active interval in the hypothesis, the period $\int_{\gamma_t} \alpha_t$ vanishes and hence, by analytic continuation, on every continuation of that branch. Therefore $[\alpha_t]$ annihilates

$$\text{span}\{g\gamma_t - \gamma_t : g \in G\}.$$

By Corollary 6.3, this is the complete projective weight-one homology. The one-place-at-infinity hypothesis identifies this with $H_1(C_t)$ of the affine fibre. The de Rham–Betti pairing is perfect, so $[\alpha_t] = 0$. $\square$

Lemma 7.6 (The primitive one-form is fibrewise exact). *Under the hypotheses of Lemma 7.5, the restriction of ω to the generic fibre is exact.*

Proof. Equation (7.2) and Lemma 7.5 show that $[\omega_t]$ is horizontal. It is therefore monodromy invariant. For a Picard–Lefschetz transvection, invariance of the cohomology class forces its period on the corresponding vanishing cycle to be zero. Since the vanishing cycles span $H_1(C_t)$, all periods of ω_t vanish. Hence the restriction is exact. $\square$

Theorem 7.7 (Polynomial relative Stokes). *Let (D, q, Ω) be admissible observable. If the phase density vanishes on every regular interval, there is $a \in k[x, y]$ such that*

$$\Omega = da \wedge dq. \quad (7.3)$$

Consequently

$$e^{zq}\Omega = d(e^{zq}a \, dq). \quad (7.4)$$

Proof. By (A5) there is an active regular interval with nonzero Lefschetz incidence. The density vanishes there by hypothesis, so Lemmas 7.5 and 7.6 show that ω is topologically relatively exact. Apply Theorem 7.4:

$$\omega = dR + a \, dq$$

with $R, a \in \mathbb{C}[x, y]$. Taking exterior derivatives and using (7.1) gives (7.3). The coefficients descend to k : after fixing the finitely many monomials occurring in one solution, coefficient comparison is a finite linear system over k . Finally, $dq \wedge dq = 0$ gives (7.4). $\square$

Proof of Theorem 7.2. Functional vanishing and Lemma 2.3 make every regular phase density zero. Apply Theorem 7.7 and ordinary Stokes:

$$F(z) = \int_{\partial D} e^{zq} a \, dq.$$

Every face is polynomially parametrized, so the right side is a finite combination of compact polynomial-line functions. It vanishes identically because $F \equiv 0$. The functional formal theorem of [13] makes this boundary relation formal. $\square$

Remark 7.8 (Why one place at infinity appears). If the generic affine fibre has $r > 1$ points at infinity, then $H_1(C_t)$ contains an additional rank- $r-1$ puncture lattice. Projective Picard–Lefschetz observability does not detect it. One may replace (A2) by an explicit infinity-lattice observability hypothesis, but it cannot simply be omitted.

8 Hodge rigidity and Jacobian correspondences

Theorem 8.1 (Deligne uniqueness for irreducible factors). *An irreducible complex local system underlying a polarizable complex variation of Hodge structure carries such a variation uniquely up to tensoring by a rank-one complex Hodge structure $\mathbb{C}(r, s)$, equivalently up to a shift of the Hodge bigrading.*

This is [6, Proposition 1.13]; see also the convenient formulation in [1, Theorem 2].

Theorem 8.2 (Weight-one Hodge rigidity). *Let $\mathbb{V}$ and $\mathbb{W}$ be nonzero polarizable rational variations of Hodge structure of effective weight one on a connected smooth algebraic base. Assume their complex local systems are irreducible. Every nonzero flat map*

$$\phi : \mathbb{V}_{\mathbb{C}} \longrightarrow \mathbb{W}_{\mathbb{C}}$$

is a morphism of complex variations of Hodge structure.

Proof. Irreducibility makes ϕ an isomorphism of complex local systems. Transport the Hodge bigrading of $\mathbb{W}$ to $\mathbb{V}$. By Theorem 8.1, the transported grading differs from the original one by a shift $\mathbb{C}(r, s)$. Because both variations are effective of weight one and arise from rational Hodge structures, both types $(1, 0)$ and $(0, 1)$ occur. The shifted set of types can again be exactly $\{(1, 0), (0, 1)\}$ only when $r = s = 0$. Thus ϕ preserves the Hodge bigrading and hence the Hodge filtration. $\square$

Remark 8.3 (The reducible warning). The theorem fails for repeated reducible local systems. A constant elliptic curve gives a two-dimensional trivial local system; most constant matrices are flat but do not preserve $H^{1,0}$. The irreducibility hypothesis is therefore load-bearing.

Proposition 8.4 (Algebraic realization). *Let $f_i : \mathcal{C}_i \rightarrow S$ and $f_j : \mathcal{C}_j \rightarrow S$ be smooth proper curve families over a smooth connected complex algebraic base, with relative Jacobians J_i, J_j . A rational morphism of weight-one variations*

$$R^1 f_{i*} \mathbb{Q} \longrightarrow R^1 f_{j*} \mathbb{Q}$$

is induced, after multiplication by a positive integer and with the usual contravariance convention, by a homomorphism of abelian schemes between J_i and J_j . Its action on R^1 is represented by an algebraic divisor correspondence on $\mathcal{C}_i \times_S \mathcal{C}_j$.

Proof. The Hodge realization functor from abelian schemes over S to torsion-free polarizable integral variations of types $(-1, 0)$ and $(0, -1)$ is fully faithful; see [5, Rappel (4.4.3)]. After multiplying the rational morphism by an integer, full faithfulness therefore supplies the corresponding homomorphism of relative Jacobians.

The homomorphism and the Poincaré bundle determine a relative line bundle on $J_i \times_S J_j$. Pulling it back along the Abel–Jacobi maps (after an inessential finite base change if relative degree-one divisor classes are needed) gives a line bundle on $\mathcal{C}_i \times_S \mathcal{C}_j$. Its first Chern class, represented by a relative divisor with rational coefficients after rescaling, acts on R^1 as the original morphism. $\square$

Corollary 8.5 (Complex intertwiners are generated by correspondences). *Under the hypotheses of Theorem 8.2, every complex flat intertwiner is a complex linear combination of rational VHS morphisms and therefore, by Proposition 8.4, of algebraic Jacobian correspondences.*

Proof. For rational local systems, the intertwiner space is the kernel of a finite system of rational linear equations. Hence scalar extension gives

$$\mathrm{Hom}_{\pi_1}(\mathbb{V}, \mathbb{W}) \otimes_{\mathbb{Q}} \mathbb{C} \simeq \mathrm{Hom}_{\pi_1}(\mathbb{V}_{\mathbb{C}}, \mathbb{W}_{\mathbb{C}}).$$

Apply Theorem 8.2 and Proposition 8.4 to the rational summands. $\square$

9 Multiphase functional comparison

Let S be the phase line with all critical and boundary branch values removed. After a common finite étale base change, assume that all face points are sections. For each datum let

$$0 \rightarrow V_j \rightarrow H_j^{\mathrm{rel}} \rightarrow \mathcal{F}_j \rightarrow 0$$

be the relative-homology sequence and let γ_j be the sliced relative class. Let α_j denote the Gelfand–Leray cohomology class.

9.1 The direct-sum variation space

Set

$$V = \bigoplus_{j=1}^N V_j, \quad \Gamma = (\gamma_1, \dots, \gamma_N),$$

and

$$\mathcal{O}_\Gamma = \text{span}\{g\Gamma - \Gamma : g \in \pi_1(S)\} \subseteq V. \quad (9.1)$$

The projection of $\mathcal{O}_\Gamma$ to V_j is $\mathcal{O}_{\gamma_j}$ and hence, for an observable datum, is all of V_j .

Lemma 9.1 (Disjoint irreducible factors). *Let $V_1, \dots, V_N$ be pairwise nonisomorphic irreducible complex representations of a group G . If a G -submodule $W \subseteq \bigoplus_j V_j$ projects nontrivially to every V_j , then*

$$W = \bigoplus_j V_j.$$

Proof. The finite-dimensional image algebra of $\mathbb{C}[G]$ is semisimple on the direct sum. A submodule is therefore a direct sum of some of the pairwise nonisomorphic simple factors. Nonzero projection to each factor forces every factor to occur. $\square$

9.2 Proof in the disjoint-block case

Theorem 9.2 (Disjoint-block functional comparison). *Let (D_j, q_j, Ω_j) be admissible observable data. On the chosen common regular phase cover labeling their face points, suppose that the systems $V_{j,\mathbb{C}}$ are irreducible, pairwise nonisomorphic, and that every active slice remains observable. Every relation*

$$\sum_j c_j \int_{D_j} e^{zq_j} \Omega_j \equiv 0, \quad c_j \in k,$$

is generated by linearity, polynomial relative Stokes, and compact polynomial-line functional relations on the faces. The hypotheses concern the representations on this common cover, not only on an earlier base.

Proof. By Lemma 2.4, the combined phase density vanishes on every open regular interval. Fix such an interval e and let J_e be the indices for which the sliced chain is nonzero on e . Analytic continuation to a base point of the common regular phase line gives, for every $g \in \pi_1(S)$,

$$\sum_{j \in J_e} c_j \langle \alpha_j, g\gamma_{j,e} \rangle = 0.$$

Subtract the equation for $g = 1$. The covector

$$\alpha_e = (c_j \alpha_j)_{j \in J_e} \in \left(\bigoplus_{j \in J_e} V_j \right)^\vee$$

annihilates the orbit-variation submodule associated with the tuple $(\gamma_{j,e})_{j \in J_e}$. By admissibility, its projection to every V_j , $j \in J_e$, is surjective. The factors remain pairwise nonisomorphic, so Lemma 9.1 makes the orbit submodule the whole direct sum. Hence $c_j \alpha_j = 0$ for every $j \in J_e$.

Every nonzero two-chain term is active on at least one regular interval. Repeating the argument over the finite interval decomposition therefore gives $c_j \alpha_j = 0$ for every index with $c_j \neq 0$. The proof of Theorem 7.7 then applies to the scaled form $c_j \Omega_j$ and gives a polynomial a_j with

$$c_j \Omega_j = da_j \wedge dq_j.$$

Stokes reduces the relation to the faces. The face terms are compact polynomial-line functional relations; they are generated by the formal operations of [13]. Their zero-dimensional endpoint terms are already part of the compact polynomial-line formal space. $\square$

Remark 9.3 (What a Jacobian correspondence does not supply). A rational weight-one morphism has the algebraic realization above, but a complex scalar multiple need not become rational after multiplication by an integer. For example, $\sqrt{2}\text{id}$ never does. Moreover, an isomorphism of generic absolute systems does not identify actual relative slice classes or preserve a polynomial lattice in their de Rham representatives. The general isotypic formal comparison is not asserted here without those additional comparisons. None of them is assumed in the elliptic theorem below.

Part II

Actual boundary systems and elliptic comparison

10 Polynomial representatives and literal boundary forcing

Put $k = \overline{\mathbb{Q}}$ and $F_0 = k(z)$. Let D be a finite algebraic-linear combination of pushforwards of $[0, 1]^2$ by polynomial maps with real algebraic coefficients. Let $q \in k[x, y]$. For a polynomial two-form ω , write

$$I_D(\omega) = \int_D e^{zq} \omega.$$

When coefficients of ω lie in F_0 , interpret this as a meromorphic function of z by linearity. For each oriented face parametrization $\phi_e : [a_e, b_e] \rightarrow \mathbb{A}^2$, put $p_e = q \circ \phi_e$. The actual boundary field $K_{\partial,0}$ is the differential field over F_0 generated by all functions

$$\int_{a_e}^{b_e} h(t) e^{z p_e(t)} dt, \quad h \in k[t],$$

and the exponential functions at their algebraic endpoints. Only the fixed faces are used; this does not mean the field of all compact line periods.

Let d differentiate only in x, y , and set

$$d_z = d + z dq \wedge, \quad H_q^2 = \Omega_{F_0[x,y]/F_0}^2 / d_z \Omega_{F_0[x,y]/F_0}^1.$$

Theorem 10.1 (Polynomial-representative boundary system). *The space H_q^2 is finite dimensional. There are forms $\omega_i = r_i(x, y) dx \wedge dy$, with $r_i \in k[x, y]$, whose classes form an F_0 -basis, a matrix $A \in M_n(F_0)$, and polynomial-in- x, y one-forms $\beta_i \in \Omega_{F_0[x,y]/F_0}^1$ such that*

$$q\omega_i = \sum_j A_{ij}(z) \omega_j + d_z \beta_i. \tag{10.1}$$

The actual integral column $U_i = I_D(\omega_i)$ satisfies

$$U' = AU + b, \quad b_i = \int_{\partial D} e^{zq} \beta_i \in K_{\partial,0}. \tag{10.2}$$

Every polynomial two-form ω has an identity

$$\omega = \sum_i a_i(z)\omega_i + d_z\beta, \quad (10.3)$$

where $a_i \in F_0$ and β is polynomial in x, y . Consequently

$$I_D(\omega) = \sum_i a_i(z)U_i + \int_{\partial D} e^{zq}\beta. \quad (10.4)$$

For any finite set of amplitudes, these functions belong to a finite closed rational first-order system whose coordinates are actual E -functions. No spatial denominator is introduced. Parameter denominators may occur.

Proof. Algebraic direct image of the rank-one connection $d + d(zq)$ on $\mathbb{A}^2 \times \mathbb{A}_z^1$ is holonomic. The generic top relative de Rham cohomology is therefore finite dimensional over $k(z)$. Equivalently, it is the generic localization of the Fourier transform of the polynomial Gauss–Manin system. The polynomial complex and Fourier convention are spelled out in [17, Proposition 9.1 and Remarks 9.4], with Fourier parameter $\tau = -z$. This finiteness statement does not identify the rank with a Milnor number without an additional hypothesis.

The monomial two-forms with coefficients in k span the quotient over F_0 , so a finite subset is a basis. The connection on the quotient is

$$\nabla_z[\omega] = [\partial_z\omega + q\omega],$$

because

$$[\partial_z + q, d + z dq\wedge] = dq \wedge - dq\wedge = 0.$$

Expressing $q\omega_i$ in the basis proves (10.1); expressing an arbitrary class proves (10.3).

For a one-form polynomial in the spatial variables, there is the literal identity

$$e^{zq}d_z\beta = d_{x,y}(e^{zq}\beta).$$

Stokes on the original parametrized compact chain proves (10.2) and (10.4). Each pullback of β to a face is a finite F_0 -linear combination of polynomial one-forms in its parameter. Thus every forcing term is an actual face integral with coefficients in F_0 .

To see finiteness on the boundary, if $\deg p = d \geq 2$, division by $\partial_t + zp'(t)$ reduces every polynomial amplitude to degree at most $d - 2$. The leading degree of $(\partial_t + zp')h$, for $h \neq 0$, is $\deg h + d - 1$, with nonzero leading coefficient in F_0 . The remainder is therefore unique. Integration of the exact part gives only endpoint exponentials. Multiplication by p and the same division give a rational first-order system for the reduced moments and the endpoint exponentials. For $d \leq 1$ every polynomial amplitude reduces directly to endpoint terms. For finitely many faces, this supplies a finite boundary vector; its fraction field is $K_{\partial,0}$.

The integral coordinates are entire and have algebraic Taylor coefficients. After pullback to the square they have the form $\int_{[0,1]^2} P(u,v)e^{zQ(u,v)} du dv$. For the first m Taylor coefficients, one denominator of the form

$$d^{m+1} \text{lcm}(1, \dots, Cm + C_0)^2$$

suffices, with fixed integers d, C, C_0 ; all conjugates grow exponentially. The same argument in one variable handles the faces. Together with the constructed differential systems, these are the E -function conditions. $\square$

Proposition 10.2 (Actual affine comparison under a change of representatives). *Suppose $\tilde{\omega} = T(z)\omega + d_z\gamma$, with $T \in \mathrm{GL}_n(F_0)$. Put $g = \int_{\partial D} e^{zq}\gamma$. Then the actual columns satisfy*

$$\tilde{U} = TU + g,$$

and their systems are related by

$$\tilde{A} = (T' + TA)T^{-1}, \quad \tilde{b} = Tb + g' - \tilde{A}g.$$

In particular, this is a comparison of the actual affine solution objects, not a statement only about their homogeneous composition factors.

Proof. The first equality is Stokes, and differentiating it proves the other two. If two primitives give the same form identity, their difference is d_z -closed and its integral over ∂D is zero by the same Stokes identity. Thus primitive choices do not introduce an arbitrary constant. $\square$

After adjoining complex constants, write $F = \mathbb{C}(z)$, $K_\partial = \mathbb{C}K_{\partial,0}$, and work in a field $\mathcal{M}$ of meromorphic germs containing the actual functions. Integration modulo the actual boundary field gives a canonical differential map

$$K_\partial \otimes_{F_0} H_q^2 \longrightarrow \mathcal{M}/K_\partial, \quad [\omega] \longmapsto I_D(\omega) \bmod K_\partial. \quad (10.5)$$

Here $\mathcal{M}/K_\partial$ is used as a differential vector space, not as a quotient field or ring. This map need not be injective for arbitrary chains. The next criterion proves injectivity rather than building it into a completion definition.

11 An actual-solution comparison criterion

Let $\mathcal{B}$ be a finite rational boundary differential system, let L_∂/F be a Picard–Vessiot field for that system containing its actual solution coordinates, and let $\mathcal{N} \subset L_\partial$ be their finite-dimensional F -linear differential span. Include 1 when needed. Let $\mathcal{T}_\partial$ be the tensor category generated by $\mathcal{B}$, allowing subquotients and duals. Its objects are precisely the differential modules trivialized by L_∂ [18, Theorem 1.5.2].

Lemma 11.1 (Differentially finite elements of a Picard–Vessiot field). *Every finite-dimensional F -differential subspace of L_∂ belongs to $\mathcal{T}_\partial$.*

Proof. Let $y \in L_\partial$ satisfy a scalar differential equation over F . Every Galois conjugate of y satisfies the same equation. Since there are no new constants, their constant span V is finite dimensional. Choose a basis $y_1, \dots, y_r$ of V . Its Wronskian is nonzero, and

$$\mathcal{L}_V(f) = \frac{\mathrm{Wr}(y_1, \dots, y_r, f)}{\mathrm{Wr}(y_1, \dots, y_r)}$$

is a monic scalar operator of order r . Its coefficients lie in L_∂ and are fixed by its entire Galois group: a basis change of V multiplies numerator and denominator by the same nonzero constant. Thus $\mathcal{L}_V \in F[\partial_z]$. It has a full fundamental set in L_∂ , so its differential module is trivialized there. The F -differential span of y is a quotient of its coefficient module and belongs to $\mathcal{T}_\partial$. Applying this construction to finitely many basis elements proves the assertion, since the category is closed under finite direct sums and quotients. See [18, Theorems 1.3.9 and 1.5.2] for the fixed-field and tensor correspondences. $\square$

Theorem 11.2 (Actual one-column comparison). *Suppose an actual column U satisfies $U' = AU + b$, with $b_i \in \mathcal{N}$, and $n \geq 2$. Assume:*

1. *the homogeneous Galois group over F contains SL_n ;*
2. *its adjoint module $\mathrm{End}^0(A)$ does not belong to $\mathcal{T}_\partial$;*
3. *at least one coordinate of U is not in $\mathcal{N}$.*

Then $U_1, \dots, U_n$ are algebraically independent over $K_\partial \subset L_\partial$. In the setting of Theorem 10.1, (10.5) is injective, and its image is the span of the actual polynomial-amplitude periods modulo K_∂ .

Proof. Let $E = \mathcal{N} + \sum_i F U_i$. The differential module $E/\mathcal{N}$ is a quotient of the coefficient module associated with A . This module is irreducible because the homogeneous group contains SL_n ; taking a dual according to the convention for connection matrices does not affect the argument. By the third assumption, the quotient is nonzero and hence is that entire irreducible module.

If all U_i belonged to L_∂ , Lemma 11.1 would put E , then $E/\mathcal{N}$, and then its adjoint module in $\mathcal{T}_\partial$, contradicting the second assumption. Thus $U \notin L_\partial^n$.

Choose the joint Picard–Vessiot field $L_h L_\partial / F$ inside a common no-new-constants differential overfield. Restriction onto the homogeneous group G is surjective. The kernel of restriction to L_∂ therefore maps to a closed normal subgroup $H \triangleleft G$. It is the homogeneous group after base change to L_∂ .

Here is the needed normal-subgroup dichotomy, including finite components. Since $G \subset \mathrm{GL}_n$ contains SL_n , its derived subgroup is SL_n . The normal subgroup $H \cap \mathrm{SL}_n$ is either all of SL_n or is finite central. In the latter case, for fixed $h \in H$ the commutator morphism $\mathrm{SL}_n \rightarrow H \cap \mathrm{SL}_n$, $s \mapsto [h, s]$, is constant by connectedness and is the identity at $s = I$. Thus h centralizes SL_n and is scalar. Hence either $H \supset \mathrm{SL}_n$ or H acts trivially on $\mathrm{End}^0(A)$. The latter would trivialize the adjoint module over L_∂ , contrary to the second assumption. It follows that the homogeneous group over L_∂ , and hence over K_∂ , contains SL_n . Every subgroup of GL_n containing SL_n is reductive: its unipotent radical intersects SL_n trivially and has trivial image in the multiplicative determinant quotient, hence is trivial.

Let Y be a fundamental matrix over a joint field for the inhomogeneous system over K_∂ . Its affine Galois group is a subgroup of $\mathbb{C}^n \rtimes G_{K_\partial}$, acting by $\sigma(U) = U + Y c_\sigma$. Its translation kernel is an invariant vector subspace, hence either 0 or $\mathbb{C}^n$. If it is $\mathbb{C}^n$, translations make the orbit of U Zariski dense in $\mathbb{A}^n$. If it is zero, the remaining regular algebraic cocycle of the reductive homogeneous group is a coboundary. Thus

$$U = a + Y c, \quad a \in K_\partial^n, \quad c \in \mathbb{C}^n.$$

The vector c is nonzero, since otherwise $U \in K_\partial^n$. The SL_n -orbit of a nonzero vector is $\mathbb{A}^n \setminus \{0\}$, again dense. In either case a polynomial over K_∂ vanishing at U vanishes on a dense affine orbit and is zero. Linear independence modulo the boundary field, and then the last assertion by normal-form reduction, follow immediately. $\square$

Remark 11.3. The third condition concerns only rational *linear* combinations of a finite actual boundary vector. It is strictly weaker than the desired nonmembership in the boundary fraction field. The proof is what upgrades this linear test to the fraction-field statement. A nonzero abstract cohomology class alone would not suffice.

12 The elliptic polynomial normal form

Fix the counterclockwise triangle

$$D = \text{conv}\{(-1/2, -1), (1/2, 0), (-1/2, 1)\} = \{-1/2 \leq x \leq 1/2, |y| \leq 1/2 - x\},$$

and

$$q = y^2 - x^3 + 3x, \quad M_j = \int_D x^j e^{zq} dx dy \quad (j = 0, 1).$$

A polynomial parametrization is $\Phi(u, v) = (u - 1/2, (1 - u)(2v - 1))$, with Jacobian $2(1 - u)$.

For $\beta = P dy - Q dx$, one has

$$d_z \beta = (L_x P + L_y Q) dx \wedge dy, \quad L_x = \partial_x + 3z(1 - x^2), \quad L_y = \partial_y + 2zy.$$

The operators act in separate variables. Their one-variable quotients have bases $1, x$ and 1 , respectively; both operators are injective on polynomials. Tensoring the two one-variable complexes proves

$$H_q^2 = F_0[dx \wedge dy] \oplus F_0[x dx \wedge dy]. \quad (12.1)$$

In particular the rank is exactly two, not merely at most two.

The explicit reduction recurrences are

$$[x^i y^j] = -\frac{j-1}{2z} [x^i y^{j-2}] \quad (j \geq 1), \quad (12.2)$$

$$[x^{m+2}] = [x^m] + \frac{m}{3z} [x^{m-1}] \quad (m \geq 0), \quad (12.3)$$

with the zero-coefficient terms omitted. Each use records a polynomial primitive. Hence every $r \in k[x, y]$ has a unique remainder $a_r(z) + b_r(z)x$, with $a_r, b_r \in k[z, z^{-1}]$, and an identity

$$r = a_r + b_r x + L_x P_r + L_y Q_r, \quad P_r, Q_r \in k[z, z^{-1}, x, y]. \quad (12.4)$$

Let

$$C = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -5/6 & 0 \\ 0 & -7/6 \end{pmatrix}, \quad A = C + D_0/z.$$

The following are identities of polynomial forms, with Laurent coefficients in z :

$$q dx \wedge dy = -\frac{5}{6z} dx \wedge dy + 2x dx \wedge dy + d_z \beta_0, \quad (12.5)$$

$$qx dx \wedge dy = 2 dx \wedge dy - \frac{7}{6z} x dx \wedge dy + d_z \beta_1, \quad (12.6)$$

where

$$\beta_0 = \frac{x}{3z} dy - \frac{y}{2z} dx, \quad \beta_1 = \frac{x^2 - 2}{3z} dy - \frac{xy}{2z} dx.$$

Define actual face integrals

$$B_{x,j} = \int_{\partial D} x^j e^{zq} dy, \quad B_{y,j} = - \int_{\partial D} x^j y e^{zq} dx.$$

Then

$$\begin{pmatrix} M_0 \\ M_1 \end{pmatrix}' = \begin{pmatrix} -5/(6z) & 2 \\ 2 & -7/(6z) \end{pmatrix} \begin{pmatrix} M_0 \\ M_1 \end{pmatrix} + \frac{1}{z} \begin{pmatrix} B_{y,0}/2 + B_{x,1}/3 \\ B_{y,1}/2 + B_{x,2}/3 - 2B_{x,0}/3 \end{pmatrix}. \quad (12.7)$$

This is obtained by integrating (12.6), not by choosing an isomorphic abstract connection and then normalizing a solution.

12.1 A finite actual boundary field

Put

$$p(x) = -x^3 + x^2 + 2x + \frac{1}{4}, \quad P_j(z) = \int_{-1/2}^{1/2} x^j e^{zp(x)} dx, \quad Q_0(z) = \int_{-1}^1 e^{z(y^2 - 11/8)} dy,$$

and $E_a = e^{-3z/8}$, $E_b = e^{11z/8}$. Parametrizing the three oriented edges gives

$$B_{x,j} = 2P_j - (-1/2)^j Q_0, \quad B_{y,j} = P_j - 2P_{j+1}. \quad (12.8)$$

The one-variable reductions imply

$$K_{\partial,0} = k(z)(P_0, P_1, Q_0, E_a, E_b). \quad (12.9)$$

Indeed the two slanted faces have the same phase p ; their arbitrary polynomial amplitudes reduce to P_0, P_1 and endpoints. The vertical quadratic amplitudes reduce to Q_0 and endpoints.

For completeness, a closed boundary system is

$$P'_0 = \left(\frac{17}{36} - \frac{1}{3z}\right) P_0 + \frac{14}{9} P_1 + \frac{E_b + 5E_a}{18z}, \quad (12.10)$$

$$P'_1 = \left(\frac{28}{27} + \frac{1}{9z}\right) P_0 + \left(\frac{163}{108} - \frac{2}{3z}\right) P_1 + \frac{-53E_b + 41E_a}{108z}, \quad (12.11)$$

$$Q'_0 = \left(-\frac{11}{8} - \frac{1}{2z}\right) Q_0 + \frac{E_a}{z}, \quad E'_a = -\frac{3}{8} E_a, \quad E'_b = \frac{11}{8} E_b. \quad (12.12)$$

For example, $P_2 = (2P_1 + 2P_0)/3 - (E_b - E_a)/(3z)$, so (12.7) closes with this boundary vector. The only finite pole of the combined rational system is zero.

12.2 The actual normalization at zero

The face forcing in (12.7) is $b(z)/z$ with b entire. Writing $M = \sum v_m z^m$, $b = \sum b_m z^m$, its holomorphic solution obeys

$$(mI - D_0)v_m = Cv_{m-1} + b_m, \quad v_{-1} = 0.$$

Every $mI - D_0$, $m \geq 0$, is invertible. Thus there is at most one holomorphic solution at zero with these actual forcing functions. The integral supplies it, with

$$M_0(0) = 1, \quad M_1(0) = -\frac{1}{6}.$$

This supplies a second, independent check that no unspecified homogeneous solution has been added to the actual column.

13 Explicit comparison with elliptic Gauss–Manin periods

The regular fibers are

$$E_t : y^2 = f_t(x) = x^3 - 3x + t.$$

On a locally constant closed cycle, let $h_j(t) = \int x^j dx/(2y)$ for $j = 0, 1$. Set

$$R_0(x, t) = \frac{4 - tx - 2x^2}{6(t^2 - 4)}, \quad R_1(x, t) = \frac{2t - 2x - tx^2}{6(t^2 - 4)}.$$

Direct differentiation of rational forms gives

$$\partial_t \left(\frac{x^j dx}{2y} \right) - \sum_{k=0}^1 G_{jk}(t) \frac{x^k dx}{2y} = d_x \left(\frac{R_j(x, t)}{y} \right), \quad G(t) = \frac{1}{6(t^2 - 4)} \begin{pmatrix} -t & 2 \\ -2 & t \end{pmatrix}. \quad (13.1)$$

These identities can be checked after multiplication by $4y^3$. Integration on closed cycles proves

$$h' = G(t)h, \quad (tI - C)h' = -(D_0 + I)h. \quad (13.2)$$

The forms $dx/(2y), x dx/(2y)$ are the usual two-dimensional de Rham basis of the elliptic curve (the latter is of the second kind at infinity). The exact rational-form identities identify the actual Gauss–Manin connection, not only its local exponents.

With Laplace convention $\mathcal{L}(tT) = \partial_z \mathcal{L}(T)$ and $\mathcal{L}(\partial_t T) = -z\mathcal{L}(T)$, applying the Weyl substitution to (13.2) gives

$$-\partial_z(zU) - C(-zU) = -(D_0 + I)U, \quad \text{hence} \quad zU' = zCU + D_0U.$$

This is exactly the homogeneous part of (12.7). The polynomial certificates (12.6) give its actual relative forcing. These two calculations identify the displayed Weyl presentation and the actual polynomial relative forcing. A connection on the punctured phase line alone would not specify its extension across the nodes; no uniqueness of an unspecified holonomic completion is being asserted or used. The independence argument below uses the explicit matrix (12.7) and the actual density, not such a uniqueness claim.

14 The actual column is not linearly generated by the faces

In this section all linear coefficients may be complex. Let

$$\mathcal{N} = \text{span}_{\mathbb{C}(z)} \{1, P_0, P_1, Q_0, E_a, E_b\}.$$

It is a differential space by the displayed boundary system.

Lemma 14.1 (Nonzero actual quotient). *One has $M_0 \notin \mathcal{N}$.*

Proof. Suppose otherwise. Clearing rational denominators yields

$$P(z)M_0(z) = \sum_j R_j(z)g_j(z), \quad 0 \neq P \in \mathbb{C}[z], \quad R_j \in \mathbb{C}[z], \quad g_j \in \{1, P_0, P_1, Q_0, E_a, E_b\}.$$

The inverse Laplace distributions of the actual line periods have algebraic densities on every regular real interval: on a monotone branch of a polynomial phase the density is $h(x(t))/p'(x(t))$. The exponentials and 1 give point masses. Derivatives of these distributions still have algebraic germs on regular intervals. Laplace uniqueness for compactly supported distributions therefore implies that the density germ $\rho(t)$ of M_0 , on a regular interval just to the right of $t = 0$, satisfies

$$P(-\partial_t)\rho(t) \text{ is algebraic over } \mathbb{C}(t). \quad (14.1)$$

We verify a nonzero elliptic monodromy variation of this particular germ. Put $f(x) = -x^3 + 3x$, let $r_2(s) \in (-1, 1)$ be its middle inverse branch for $-2 < s < 2$, and let $c(t)$ be the inverse branch of p beginning at $c(0) \in (-1/5, 0)$. Set $a(t) = 1/2 - c(t)$. Coarea on the triangle gives, near zero,

$$\rho(t) = \int_{-a(t)}^{a(t)} \frac{dy}{3(1 - r_2(t - y^2)^2)}. \quad (14.2)$$

We check the continuation and its normalization explicitly. On $[-1/5, 1]$ one has $p' > 0$, with

$$p(-1/5) = -51/500 < 0 < p(0) = 1/4, \quad p(4/5) = 989/500 < 2 < p(1) = 9/4.$$

Thus the branch $c(t)$ is analytic on $0 \leq t \leq 2$, $c(0) \in (-1/5, 0)$, and $c(2) \in (4/5, 1)$. For $0 < t < 2$ and $|y| \leq |a(t)|$,

$$t - y^2 \in [f(c(t)), t] \subset (-2, 2), \quad f(c(t)) = t - a(t)^2.$$

Consequently the inverse branch r_2 in (14.2) is unambiguous throughout this real continuation. At $t_* = 11/8$ the endpoints cross because $c(t_*) = 1/2$. The expression

$$a(t) \int_{-1}^1 \frac{dv}{3(1 - r_2(t - a(t)^2 v^2)^2)}$$

is holomorphic near t_* , proving that the crossing does not obstruct analytic continuation. For $t > t_*$ its orientation is reversed.

At $t = 2$ the endpoints remain distinct and unramified, since $a(2) < -3/10$ and $p'(c(2)) \neq 0$. Put $\varepsilon = 2 - t$ and $r_2(t - y^2) = 1 - v$. The identity

$$f(1 - v) = 2 - 3v^2 + v^3$$

gives, uniformly near the node on the chosen branch,

$$\frac{1}{3(1 - r_2(t - y^2)^2)} = \frac{1}{2\sqrt{3}\sqrt{\varepsilon + y^2}} + O(1).$$

Splitting the integral at a small fixed neighborhood of $y = 0$, and using the reversed orientation, yields the nonzero coefficient

$$\rho(2 - \varepsilon) = \frac{1}{2\sqrt{3}} \log \varepsilon + O(1), \quad \varepsilon \downarrow 0. \quad (14.3)$$

Outside that neighborhood the relative path and its endpoints transport holomorphically around a small loop about 2. Inside it the nodal neck is an annulus. Its monodromy changes a crossing relative path by a multiple of its core circle; this is the local Picard–Lefschetz description [8, Chapters 2–3]. The coefficient cannot vanish by (14.3). With δ_+ the core vanishing cycle,

$$(T_+ - I)\rho = c_* h_+(t), \quad c_* \neq 0, \quad h_+(t) = \int_{\delta_+} \frac{dx}{2y}, \quad T_+ h_+ = h_+.$$

The sign from $dx/(2y) = -dy/(3(1 - x^2))$ on a fiber is absorbed in c_* . This fixes the comparison with the actual oriented density and does not choose an independent normalization of a punctual distribution.

A power T_+^m fixes the algebraic germ in (14.1). Analytic continuation commutes with the constant-coefficient operator, so $P(-\partial_t)h_+ = 0$. But every solution germ of a nonzero constant-coefficient differential equation extends to an entire function of t . The elliptic period h_+ does not. Indeed, for $-2 < t < 2$ let $r_1 < r_2 < r_3$ be the roots of $x^3 - 3x + t$. Up to a fixed nonzero factor,

$$h_+(t) = \int_{r_2}^{r_3} \frac{dx}{\sqrt{(x - r_1)(x - r_2)(r_3 - x)}}.$$

As $t \rightarrow -2^+$, $r_1, r_2 \rightarrow -1$ and $r_3 \rightarrow 2$. Put $\delta = r_2 - r_1$. On a fixed small interval $x = r_2 + u$, the third factor $r_3 - x$ is bounded above and below by positive constants, and the integral is comparable to

$$\int_0^{u_0} \frac{du}{\sqrt{u(u + \delta)}} = 2 \operatorname{arsinh} \sqrt{u_0/\delta} \rightarrow \infty.$$

Thus it diverges logarithmically, contradicting entire continuation and proving the lemma. $\square$

Remark 14.2. The singularity at $t = 2$ is a singularity of the analytic continuation of a regular phase-density germ. It is not a claim that the original compactly supported density is nonzero outside $q(D) = [-11/8, 11/8]$.

15 Galois disjointness from the actual boundary system

15.1 The rank-two homogeneous group

Eliminating the second coordinate from $Y' = AY$ gives

$$y_0'' + \frac{2}{z}y_0' + \left(\frac{5}{36z^2} - 4\right)y_0 = 0.$$

With $w = zy_0$, this is

$$w'' = \left(4 - \frac{5}{36z^2}\right)w.$$

Equivalently $y_0 = z^{-1/2}v(2z)$, where v satisfies the modified Bessel equation of order $1/3$. The Bessel differential-Galois classification states that the group is SL_2 for an order not in $1/2 + \mathbb{Z}$ [18, Example 1.3.32]; an algebraic rank-one twist does not change this connected derived group. Moreover $\mathrm{tr} A = -2/z$, so the determinant solution is a constant multiple of z^{-2} . Thus the homogeneous group of our system is exactly SL_2 . Its adjoint formal exponential factors at infinity are

$$1, e^{4z}, e^{-4z}. \quad (15.1)$$

15.2 The boundary has only one nonabelian simple factor

The homogeneous rank-two block of the boundary system is the cubic moment module for p . Write

$$\alpha = \frac{\sqrt{7}}{3}, \quad \beta = \alpha^3 = \frac{7\sqrt{7}}{27}, \quad c_0 = \frac{107}{108}.$$

Then

$$p(\alpha s + 1/3) = c_0 + \beta(-s^3 + 3s). \quad (15.2)$$

The centered cubic homogeneous moments satisfy

$$V' = \left[\begin{pmatrix} 0 & 2\beta \\ 2\beta & 0 \end{pmatrix} + \frac{1}{z} \begin{pmatrix} -1/3 & 0 \\ 0 & -2/3 \end{pmatrix} \right] V,$$

up to the scalar exponential $e^{c_0 z}$ and an algebraic constant basis change. Its first coordinate is again modified Bessel of order $1/3$, now at $2\beta z$. Its derived group is SL_2 , and its adjoint formal exponential factors are

$$1, e^{4\beta z}, e^{-4\beta z}. \quad (15.3)$$

All other diagonal factors in the full boundary system are rank one: the vertical quadratic factor and the endpoint exponentials. The kernel of the Galois action on these diagonal factors is unipotent, because the system is block triangular. The reductive quotient therefore has one simple derived factor SL_2 , detected by the cubic block; its remaining quotient is diagonalizable.

Lemma 15.1 (Projective exclusion). *The adjoint module of A does not belong to the tensor category of the boundary system.*

Proof. Otherwise that boundary group would have the projective Galois group PGL_2 of A as a quotient. The unipotent radical maps trivially to this reductive quotient, so the quotient factors through the reductive boundary group. Its restriction to the unique simple derived SL_2 is the usual projective quotient up to an automorphism of PGL_2 ; such automorphisms over $\mathbb{C}$ are inner. After aligning this restriction with the cubic block, the two projective maps agree on the whole group, including disconnected components. Indeed, if S is the connected simple derived factor, both maps send $gs g^{-1}$ to conjugation of the common image of s . Their values on g therefore differ by an element centralizing PGL_2 . This centralizer is trivial. Thus the two adjoint differential modules would be isomorphic over $\mathbb{C}(z)$.

Their formal exponential factors (15.1) and (15.3) disagree, since $\beta \neq \pm 1$. Such factors are preserved by differential-module isomorphism. This is a contradiction. Rank-one endpoint exponentials do not affect an adjoint module, so they cannot remove the obstruction. $\square$

Theorem 15.2 (Complete actual elliptic comparison). *For the fixed triangle and phase above,*

$$M_0, M_1 \text{ are algebraically independent over } K_\partial.$$

In particular, over the arithmetic boundary field,

$$K_{\partial,0}[I_D(r dx \wedge dy) : r \in k[x, y]] = K_{\partial,0}[M_0, M_1] \simeq K_{\partial,0}[X_0, X_1]. \quad (15.4)$$

The canonical map (10.5) is an isomorphism onto the two-dimensional space of actual polynomial-amplitude periods modulo the boundary field. For every polynomial amplitude,

$$I_D(r dx \wedge dy) \in K_\partial \iff [r dx \wedge dy] = 0 \text{ in } H_q^2. \quad (15.5)$$

All such boundary reductions are the polynomial identities (12.4) integrated by Stokes.

Proof. The homogeneous group is SL_2 , Lemma 15.1 gives projective disjointness, and Lemma 14.1 gives the required nonzero actual quotient. Apply Theorem 11.2. The equality of algebras and the equivalence follow from the unique normal form and algebraic independence. The statement over k follows from the stronger statement after adjoining complex constants. $\square$

16 Numerical comparison and the correct fiber algebra

For $\xi \in k^\times$, put

$$K_{\partial,\xi} = k(P_0(\xi), P_1(\xi), Q_0(\xi), E_a(\xi), E_b(\xi)).$$

It is the field of all polynomial-amplitude values on the three fixed faces and their endpoints: the one-variable reductions have only powers of z in their denominators.

Theorem 16.1 (Algebraic specialization). *For every $\xi \in \overline{\mathbb{Q}}^\times$, the numbers $M_0(\xi), M_1(\xi)$ are algebraically independent over $K_{\partial,\xi}$. Consequently*

$$K_{\partial,\xi}[I_D(r dx \wedge dy)(\xi) : r \in k[x, y]] \simeq K_{\partial,\xi}[X_0, X_1].$$

Proof. All coordinates of the boundary system and of its augmentation by M_0, M_1 are E -functions, and the combined rational system has no finite singular point other than zero. Let

$$d = \mathrm{trdeg}_{k(z)} k(z)(P_0, P_1, Q_0, E_a, E_b).$$

By Theorem 15.2, adjoining M_0, M_1 raises this degree to $d + 2$. The Siegel–Shidlovskii equality, in the form recalled in [2, Theorem 1.1], applied first to the boundary vector and then to the combined vector, gives degrees d and $d + 2$ at every $\xi \neq 0$. The two additional values therefore have transcendence degree two over $K_{\partial, \xi}$. The normal-form coefficients are Laurent polynomials in z , so specialization proves the second assertion. $\square$

Corollary 16.2 (Value-level twisted exactness). *For $r \in k[x, y]$ and $\xi \in k^\times$, the following are equivalent:*

1. $I_D(r \, dx \wedge dy)(\xi)$ is algebraic over $K_{\partial, \xi}$;
2. it belongs to $K_{\partial, \xi}$;
3. $r \, dx \wedge dy \in (d + \xi \, dq \wedge) \Omega_{k[x, y]/k}^1$.

Proof. Specialize (12.4). A nonzero linear form in two algebraically independent variables is transcendental over their base field. Thus the first condition forces $a_r(\xi) = b_r(\xi) = 0$, and specialized polynomial division makes this equivalent to the third condition. Literal Stokes gives the second condition in that case. $\square$

For clarity, one can also formulate the comparison over a local ring. Let

$$R_{\partial} = k[z, P_0, P_1, Q_0, E_a, E_b], \quad \pi = z - \xi.$$

Beukers lifting [2, Theorems 1.2–1.3] gives

$$\ker(\text{ev}_{\xi} : R_{\partial} \rightarrow \mathbb{C}) = \pi R_{\partial}.$$

Indeed, a polynomial expression vanishing at ξ differs from a lifted functional relation by a multiple of π . Consequently $B_{\xi} = (R_{\partial})_{\pi R_{\partial}}$ is a DVR, and its residue field is $\kappa_{\xi} = K_{\partial, \xi}$. The local actual functional algebra is

$$B_{\xi}[M_0, M_1] \simeq B_{\xi}[X_0, X_1].$$

Its numerical kernel is exactly $\pi B_{\xi}[X_0, X_1]$, and the numerical period map is injective on the fiber

$$(B_{\xi}[X_0, X_1]) / \pi B_{\xi}[X_0, X_1] = \kappa_{\xi}[X_0, X_1].$$

There is no claim of numerical injectivity before quotienting by π .

Remark 16.3 (Formal period operations). The actual functional and numerical independence theorems above do not use an unpublished completeness theorem for the boundary. To identify all relations among boundary symbols with Kontsevich–Zagier operations, use the separate compact polynomial-line formal comparison of Paper III [14]. Then (12.4), ordinary twisted Stokes, and polynomial freeness prove that every relation for this fixed D, q and arbitrary polynomial amplitudes is generated by those boundary relations and the explicit Stokes reductions. There is no remaining determinant generator, no assumed isotypic normal-function compatibility, and no need to infer arithmetic constants from punctual support.

Part III

Local formal comparison and its scope

17 Boundary specialization without a global normal form

The local arithmetic input applies to any finite closed E -function vector; it does not require a global Laurent-polynomial presentation of its actual function algebra.

Lemma 17.1 (Direct boundary specialization). *Let $f_1, \dots, f_N$, with the constant coordinate included when needed, be a solution of a rational first-order E -function system over $k(z)$. Suppose $\xi \in k^\times$ is ordinary for that system and put*

$$R = k[z, f_1, \dots, f_N], \quad \pi = z - \xi.$$

Then $\ker(\text{ev}_\xi : R \rightarrow \mathbb{C}) = \pi R$. The localization $B = R_{\pi R}$ is a DVR, its valuation is analytic vanishing order at ξ , and

$$\kappa = B/\pi B = \text{Frac}(\text{ev}_\xi R).$$

Proof. Let $P(z, f(z)) \in R$ vanish at ξ . Beukers lifting supplies $Q(z, X) \in k[z, X]$ with $Q(z, f(z)) = 0$ and $Q(\xi, X) = P(\xi, X)$; use the constant coordinate to homogenize if necessary. Thus $P - Q = (z - \xi)H$ in $k[z, X]$ and $P(z, f(z)) = \pi H(z, f(z))$. The reverse containment is immediate.

The ring R is a finitely generated domain, and πR is a nonzero prime ideal. The principal ideal theorem gives it height one. Its localization is a noetherian local domain with nonzero principal maximal ideal, hence a DVR [19, Tag 00PD]. Every local unit has a nonzero value at ξ , so writing an element as π^m times a unit identifies the valuation with analytic order. The residue field is the fraction field of $R/\pi R$, which is the stated actual value field. $\square$

Remark 17.2 (Why no global normal form is used). The entire line function $h = (e^z - 1)/z$ gives the actual algebra

$$k[z, h, e^z, e^{-z}] = k[z, h, (1 + zh)^{-1}].$$

Here z, h are algebraically independent: otherwise $e^z = 1 + zh$ would be algebraic over $k(z)$, contrary to its essential singularity at infinity. This ring is normal but is not a polynomial extension of a Laurent polynomial algebra over $k[z]$. Its nonconstant unit $1 + zh$ becomes 1 modulo z , whereas a unit ct^a in such a Laurent-polynomial model can become 1 modulo z only if it already is 1. Reduced moment normal forms over an exponential field do not justify the stronger unlocalized claim for every actual line-function algebra.

18 An abstract local comparison theorem

Let B and κ be as in Lemma 17.1, and let $K = \text{Frac } B$. Let $U_1, \dots, U_m$ be actual E -functions which, together with the boundary vector, lie in a finite rational first-order system ordinary at ξ . Assume that an ideal $J \subset B[X_1, \dots, X_m]$ is generated by explicitly realized formal identities and satisfies

$$JK[X] = \ker(K[X] \longrightarrow K[U]), \tag{18.1}$$

$$B[X]/J \text{ is } B\text{-flat}. \tag{18.2}$$

The phrase “explicitly realized” is a separate geometric hypothesis: for example, the polynomial Stokes certificates of this paper provide such identities. It is not inferred merely from membership of their coefficients in the boundary field.

Theorem 18.1 (Functional contraction and numerical fiber). *Under (18.1)–(18.2), the functional map $B[X]/J \rightarrow \mathcal{O}_\xi^{\text{an}}$ is injective, and so is*

$$(B[X]/J) \otimes_B \kappa \longrightarrow \mathbb{C}.$$

Equivalently, the numerical kernel on $B[X]/J$ is $\pi(B[X]/J)$, not zero before specialization.

Proof. Flatness over the DVR implies torsion-freeness. Thus $B[X]/J$ injects into its generic fiber, and generic completeness gives

$$(JK[X]) \cap B[X] = J.$$

This proves the functional assertion.

For the numerical assertion, clear the finitely many boundary denominators of a representative $P \in B[X]$. Each denominator has nonzero value at ξ , so this multiplication is by a local unit. The resulting representative is $\tilde{P}(z, f(z), X)$ with $\tilde{P} \in k[z, Y, X]$. If its value at $U(\xi)$ is zero, Beukers lifting applied to the *combined* closed E -function vector gives $Q(z, Y, X)$ which vanishes functionally and satisfies

$$Q(\xi, Y, X) = \tilde{P}(\xi, Y, X).$$

After substituting $Y = f(z)$, the polynomial $Q(z, f(z), X)$ is in the functional kernel and hence in J . The difference $\tilde{P} - Q$ is divisible by π in $k[z, Y, X]$. Therefore the class of P in $B[X]/J$ belongs to $\pi(B[X]/J)$. Division by the previously cleared unit is legitimate. The reverse containment follows by evaluation, proving the numerical statement. $\square$

Definition 18.2 (Specialized formal algebra). For a local functional symbol algebra $\mathcal{P}^{\text{fun}, \text{loc}}$ over B , its numerical algebra at ξ is

$$\mathcal{P}_\xi^{\text{num}} = \mathcal{P}^{\text{fun}, \text{loc}} \otimes_B \kappa = \mathcal{P}^{\text{fun}, \text{loc}} / \pi \mathcal{P}^{\text{fun}, \text{loc}}.$$

Whenever the two hypotheses of Theorem 18.1 are proved for the stated formal generators, its numerical period map is injective. The residue field is the boundary-value field, generally not k .

19 The abstract multicolumn orbit model

The following algebraic calculation is retained for future geometric applications. Its group hypotheses and its eventual formal realization are not automatic properties of polynomial flags.

Work over a characteristic-zero differential field K with algebraically closed constants k . For finitely many blocks of ranks $n_\lambda \geq 2$, let $Y'_\lambda = A_\lambda Y_\lambda$ be fundamental matrices and let

$$U'_\lambda = A_\lambda U_\lambda + B_\lambda$$

be matrices of actual particular-solution columns in a common no-new-constants extension. Suppose the homogeneous group, in these representations, is a subgroup of $\prod_\lambda \text{GL}_{n_\lambda}$ containing $\prod_\lambda \text{SL}_{n_\lambda}$.

Proposition 19.1 (Affine orbit reduction). *Constant operations in each column multiplicity space remove the translation-active columns as independent affine coordinates. Every remaining column is split, of the form*

$$U_{\lambda, \text{sp}} = a_\lambda + Y_\lambda C_\lambda, \quad a_\lambda \in \text{Mat}(K), \quad C_\lambda \in \text{Mat}(k).$$

After recording the constant kernel of C_λ as affine-linear relations, let $s_\lambda = \text{rank } C_\lambda \leq n_\lambda$. If $s_\lambda < n_\lambda$, this block has no further polynomial relation. Only blocks with $s_\lambda = n_\lambda$ contribute determinant coordinates $\Delta_\lambda = \det(U_\lambda - a_\lambda)$.

Proof. The translation kernel is an invariant vector subgroup of the direct sum of the standard representations with their column multiplicity spaces. The product of special-linear groups acts independently, so it has the form $\bigoplus_\lambda k^{n_\lambda} \otimes R_\lambda$. Choosing constant bases of the R_λ gives independent translations in the corresponding matrix coordinates. The ideal of the solution variety is extended from the other coordinates; one does not assert that an arbitrary multiple of a relation cannot mention a translated variable.

After quotienting the translations, the affine term is a regular algebraic cocycle of the homogeneous reductive group. Complete reducibility makes it a coboundary, giving the displayed split form. Constant column operations record all kernels of C_λ . For $s < n$, the SL_n -orbit of a full-column-rank $n \times s$ matrix is the entire rank- s locus: a GL_n map can be adjusted to determinant one on a complementary vector. This locus is dense in the whole matrix space. For $s = n$, the orbit is exactly the nonzero determinant fiber. These facts give all assertions. $\square$

Let $T_{\text{det}} \subset \mathbb{G}_m^s$ be the image of the homogeneous group on the determinants of its square blocks. It may be disconnected. Put

$$L = \{a \in \mathbb{Z}^s : x^a|_{T_{\text{det}}} = 1\}, \quad \eta_a = \Delta^{a^+} / \Delta^{a^-} \in K^\times.$$

Theorem 19.2 (Generic affine relation ideal). *The complete polynomial relation ideal over K is generated by the recorded affine-linear relations and*

$$\Delta^{a^+} - \eta_a \Delta^{a^-}, \quad a \in L.$$

Proof. The affine reduction leaves full affine factors, square determinant fibers, and the orbit of the determinant vector under T_{det} . In Laurent determinant coordinates the latter is a translate of a closed subgroup of a torus, with partial-character ideal given by L . Its contraction to polynomial coordinates is the lattice ideal generated by all the displayed binomials, including when L is not saturated; see [7].

This also describes its pullback through the determinant maps, without adding a component at zero determinants. Indeed those maps are flat (Lemma 19.4), and the lattice coordinate ring embeds into its Laurent localization. Tensoring this injection with the flat matrix coordinate ring embeds the pullback into its localization at all determinants. There it is exactly the special-linear orbit family; hence its closure gives the asserted ideal. The Picard–Vessiot torsor identifies the solution relation ideal with this orbit-closure ideal. $\square$

19.1 A local model with unit normalization

Suppose now that the boundary local ring is a DVR B as above, that the block and column coordinates used in the generic theorem are obtained by invertible changes over B , and that

$$a_\lambda \in \text{Mat}(B), \quad \eta_a \in B^\times.$$

These are hypotheses on the chosen local model. For an ordinary block system they follow for an existing split expression from holomorphy of U_λ, Y_λ and invertibility of $Y_\lambda(\xi)$: its shift is $U_\lambda - Y_\lambda C_\lambda$, and the selected full-rank constant determinant is nonzero. A meromorphic block decomposition must first be replaced by a B -invertible one; this is not inferred from a generic decomposition alone.

Lemma 19.3 (Flat unit-valued lattice quotient). *For a character $\rho : L \rightarrow B^\times$, the quotient*

$$D_\rho = B[\delta_1, \dots, \delta_s] / (\delta^{a^+} - \rho(a)\delta^{a^-} : a \in L)$$

is free over B .

Proof. Declare $u, v \in \mathbb{N}^s$ equivalent if $u - v \in L$. The binomial for $u - v$, multiplied by their common monomial factor, identifies the two monomials by the unit $\rho(u - v)$. Choose one representative for every congruence class. The character identity makes all reductions consistent. Sending each monomial to the corresponding unit multiple of its chosen representative defines the inverse to the resulting free-module presentation. Hence those representatives form a B -basis. $\square$

Lemma 19.4 (Free determinant morphism). *For every $n \geq 1$, $B[x_{ij}]$ is free over $B[\delta]$ under $\delta \mapsto \det(x_{ij})$.*

Proof. It is the quotient of $B[\delta][x_{ij}]$ by $\det(x_{ij}) - \delta$. Choose a monomial order in the x_{ij} for which the diagonal product is leading. The polynomial is monic. Division by this single monic polynomial gives a unique remainder in the monomials not divisible by its leading monomial, which form a $B[\delta]$ -basis. This proof works over an arbitrary commutative coefficient ring. $\square$

Proposition 19.5 (Flat normalized affine model). *In the specified local coordinates, the quotient by the regular coordinate-linear relations and unit-valued determinant relations is B -flat and its generic ideal contracts to the same B -ideal.*

Proof. Translation by the regular shifts is a polynomial automorphism over B . After eliminating the coordinate-linear relations, the quotient is a polynomial extension of the base change of the product of determinant maps to D_ρ . The two preceding lemmas make it flat over B . Torsion-freeness then proves contraction from the generic fiber. $\square$

Remark 19.6 (Admissible clearing of denominators). Choose $\eta_a = b_a/c_a$ only with $b_a(\xi)c_a(\xi) \neq 0$. Then clearing the denominator multiplies the normalized binomial by a unit and leaves its local ideal unchanged. With $\pi = z - \xi$, the presentations $1 = 1/1 = \pi/\pi$ give $\delta - 1$ and $\pi(\delta - 1)$, which have different ideals over B . In the latter quotient a nonzero class is killed by π ; its equation disappears on specialization. Arbitrary nonzero global denominators are therefore not interchangeable in a local formal presentation.

19.2 The remaining geometric hypotheses

The preceding results prove a general numerical formal theorem only when the following data have been supplied: an actual polynomial boundary system; verified homogeneous groups over the actual boundary field; a generically complete *geometric* affine-linear comparison; a compatible local lattice; and geometric realizations of any rank-one determinant identifications. Once these give an explicit ideal J satisfying (18.1) and (18.2), Theorem 18.1 applies.

A determinant character becoming trivial in a differential Galois group identifies a rank-one differential module. It does not by itself prove that the identification is generated by pullback,

trace, and Stokes on algebraic chains. One may explicitly enlarge the formal category to admit those maps, or prove their geometric realization. The two statements should not be conflated. For the elliptic triangle, there is one actual rank-two column and the independent-coordinate ideal is zero, so this additional issue does not arise.

20 Falsification controls and remaining scope

20.1 Punctual support does not determine arithmetic normalization

For a compactly supported distribution T on the real phase line,

$$\mathcal{L}(T)(z) = \langle T, e^{zt} \rangle, \quad \mathcal{L}(\partial_t T) = -z\mathcal{L}(T), \quad \mathcal{L}(tT) = \partial_z \mathcal{L}(T).$$

Products of Laplace transforms correspond to convolution. If T is supported at a , its transform has the form $\sum_m c_m(-z)^m e^{az}$. Algebraicity of a does not imply algebraicity of the coefficients c_m .

For example, take $q = y^2 - x^3 + 3x$ near its minimum $(-1, 0)$ and a small rational rectangle containing that point. Put $u = x + 1$. Then

$$q + 2 = 3u^2 - u^3 + y^2 = X^2 + Y^2, \quad X = u\sqrt{3 - u}, \quad Y = y.$$

The area Jacobian at the origin is $1/\sqrt{3}$, so the phase density near -2 is $\mathbf{1}_{t > -2} g(t + 2)$ with $g(0) = \pi/\sqrt{3}$. Its derivative has the punctual term $(\pi/\sqrt{3})\delta_{-2}$, whose Laplace transform is $(\pi/\sqrt{3})e^{-2z}$. This is not an arithmetic E -function over k . The actual unsplit compact integral still is an E -function; the nonarithmetic constants need not survive separately in a polynomial Stokes presentation. The boundary-system theorem keeps that presentation and makes no inference from support alone.

20.2 Two elementary comparison controls

Numerical evaluation on an unspecialized local algebra kills its nonzero element $z - \xi$. This is why Definition 18.2 uses the fiber algebra. Likewise, multiplication of an ideal generator by a nonunit can preserve its generic fiber but destroy its local flatness, as the determinant countermodel above shows.

Topological relative exactness is not always polynomial relative exactness. For $q = x^2$, the form $\omega = x dy$ is exact on each component of a regular fiber. If $\omega = dR + a dq$ polynomially, its dy coefficient forces $R = xy + h(x)$; the dx coefficient would require $y + h'(x) + 2xa = 0$, impossible in $k[x, y]$. The connected-fiber and finite-critical-locus hypotheses in the geometric theorem are not cosmetic.

20.3 Verification and limits of the current theorem

The exact certificates in the elliptic proof can be checked independently by polynomial arithmetic. The repository includes a reproducible script for monomial normal forms, both actual connection identities, both Gelfand–Leray identities, the scalar equation, the full boundary system, and independently integrated Taylor coefficients. Additional rational inequalities verify the branch and endpoint conditions used in the analytic continuation. These finite checks are not a kernel-checked proof of the monodromy or differential-Galois arguments.

The boundary-system construction holds for arbitrary polynomial phases and compact polynomially parametrized two-chains. The actual independence criterion has three explicit hypotheses.

They are proved here for the fixed elliptic triangle, with arbitrary polynomial amplitudes and every nonzero algebraic specialization. The general multiphase and multicolumn geometric comparison remains subject to the hypotheses isolated in the local framework. In particular, no theorem for arbitrary isotypic relative normal functions, arbitrary boundary projective overlap, or unspecified determinant-line realizations follows from the present proof.

The real-density geometry is not used to impose a real-phase restriction on the algebraic boundary construction. For genuinely complex phases on real two-chains the phase support may be two dimensional, and the one-dimensional density-monodromy argument must be replaced; it is not silently extended to that setting.

Acknowledgments and AI-assisted research disclosure

This manuscript is part of the exponential-integral program developed with Papers I–III [12, 13, 14]. During its preparation the authors used AI assistants for proof exploration, adversarial checking, symbolic verification, bibliographic checking, and language editing. The authors are responsible for the mathematical statements, proofs, and remaining errors. The September 2026 revision replaces the earlier general endpoint-completeness claim by the explicit actual comparison and separately stated general hypotheses above.

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